

65. An astronaut whose weight on the Earth is 600 N experiences weightlessness on International Space Station orbiting around the Earth. It means that

- (a) acceleration of the astronaut is zero
- (b) normal reaction of the space-station floor on the astronaut is zero
- (c) gravitational pull of earth on the astronaut is zero
- (d) space station applies a centrifugal force on the astronaut

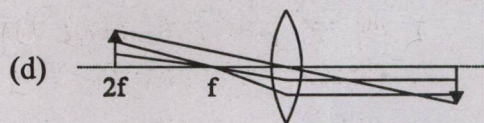
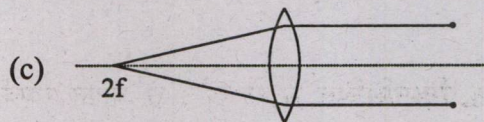
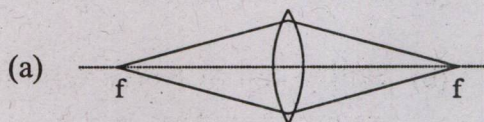
66. Which of the following statements give characteristics of contact forces ?

- 1. It appears between an object when it is in contact with some other object
- 2. It satisfies the third law of motion
- 3. It may appear between a pair of solid and fluid

Select the answer using the code given below :

- (a) 1 and 3 only
- (b) 2 and 3 only
- (c) 1 and 2 only
- (d) 1, 2 and 3

67. Which one among the following figures correctly represents the ray diagram? (Consider the lens to be thin)

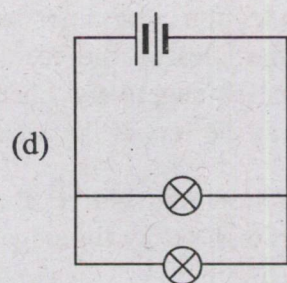
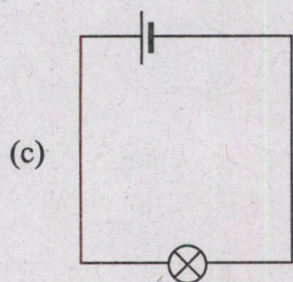
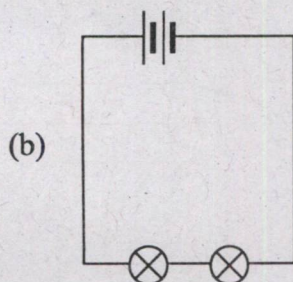
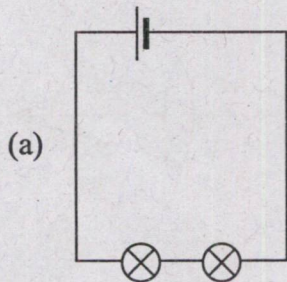


68. Lightning is due to

- (a) The flow of charges between different parts of the cloud
- (b) The short-circuiting of charges between the upper and lower surfaces of the cloud
- (c) The collection of positively charged particles on the base and collection of negatively charged particles at the top of the cloud
- (d) The induction of positive charge on the ground below the negative charge at the base of the cloud



69. निम्नलिखित में से किस स्थिति में, बल्ब  $\otimes$  सबसे अधिक उद्दीप्त होगा ? (मान लीजिए सभी बैटरियाँ एक जैसी हैं)



70. निम्नलिखित में से कौन-सा आरेख, किसी छलांग के दौरान स्काइडाइवर की गति को सटीक रूप से निरूपित करता है ?

